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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	TriHack
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 – Decent Work And Economic Growth SDG 17 – Partnership For The Goals
Project / Solution Name	Smart SMEs Procurement
Name (Leader)	Ng Xin Rou
Email & Phone (Leader)	ERA2509054@xmu.edu.my 017-272 4083
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium-sized enterprises (SMEs) often face challenges when purchasing stationery and other business supplies. Finding suitable wholesale suppliers can be time-consuming because information about product prices, shipping fees, minimum order quantities, delivery times, and supplier locations is scattered across different sources. Without a centralized comparison platform, business owners may struggle to identify suppliers that meet their budgets and operational needs.

These inefficiencies may cause higher purchasing costs, delays in obtaining essential supplies, and additional administrative work. For SMEs with limited financial and human resources, time spent manually searching for suppliers and comparing quotations could be better used to support daily operations and business growth.

Furthermore, limited access to transparent supplier information can make it difficult for smaller businesses to establish reliable partnerships with other companies. This may reduce their ability to make informed purchasing decisions and collaborate effectively within the business ecosystem.

Our project faces these challenges through an automated supplier comparison and procurement platform. Users can submit their purchasing requirements, including product quantity, budget, and desired delivery time. The system then compares supplier information, evaluates available options, and recommends the most suitable suppliers based on transparent criteria. Human approval remains part of the purchasing process to ensure accountability and control.

This initiative supports SDG 8 (Decent Work and Economic Growth) by improving SME efficiency, SDG 9 (Industry, Innovation and Infrastructure) by digitalizing procurement, SDG 12 (Responsible Consumption and Production) by encouraging responsible purchasing, and SDG 17 (Partnerships for the Goals) by strengthening partnerships between SMEs and suppliers.

(249 words)

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Smart SME Procurement is an AI-powered supplier recommendation and procurement optimization system designed to help micro, small and medium-sized enterprises (SMEs) source stationery supplies more efficiently.

Users enter their purchasing requirements, including product type, quantity, budget, and desired delivery date. The system compares wholesale suppliers based on product prices, shipping fees, minimum order quantities, delivery times, and geographical distance. A scoring algorithm evaluates these factors to rank suppliers and recommends suitable options to help SMEs make informed purchasing decisions.

The system integrates Google Gemini API to generate intelligent supplier recommendations and explain the advantages and trade-offs of different options. OSRM (Open-Source Routing Machine) supports distance and routing calculations, while GeoPy helps process geographical locations and calculate distances between SMEs and suppliers.

The website front end will be developed using Replit, while the backend will be built with Python and FastAPI to handle supplier comparisons, scoring calculations, and procurement requests. Uvicorn will serve the FastAPI application to enable the backend to receive and process API requests from the frontend. A database will store supplier information, purchase requests, and order records.

The project applies technical skills from the SDC Phase 1 workshops, including Generative AI API integration, database architecture, REST API development, and cloud hosting.

Human control remains central to the workflow. Users review supplier recommendations and select their preferred supplier before submitting a purchase request for human approval. The system records order statuses and approval outcomes to support transparency and accountability.

Smart SME Procurement supports SDG 8 by improving SME procurement efficiency, SDG 9 by digitizing purchasing workflows, SDG 12 by encouraging more informed purchasing decisions, and SDG 17 by strengthening partnerships between SMEs and suppliers.

(275 words)

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Replit is used to develop and host the website interface to allow users to enter procurement requirements, compare supplier offers, and manage purchase requests.

Backend: Python with FastAPI handles procurement requests, supplier comparison, scoring calculations, and order processing. Uvicorn serves the FastAPI application and enables communication through REST APIs.

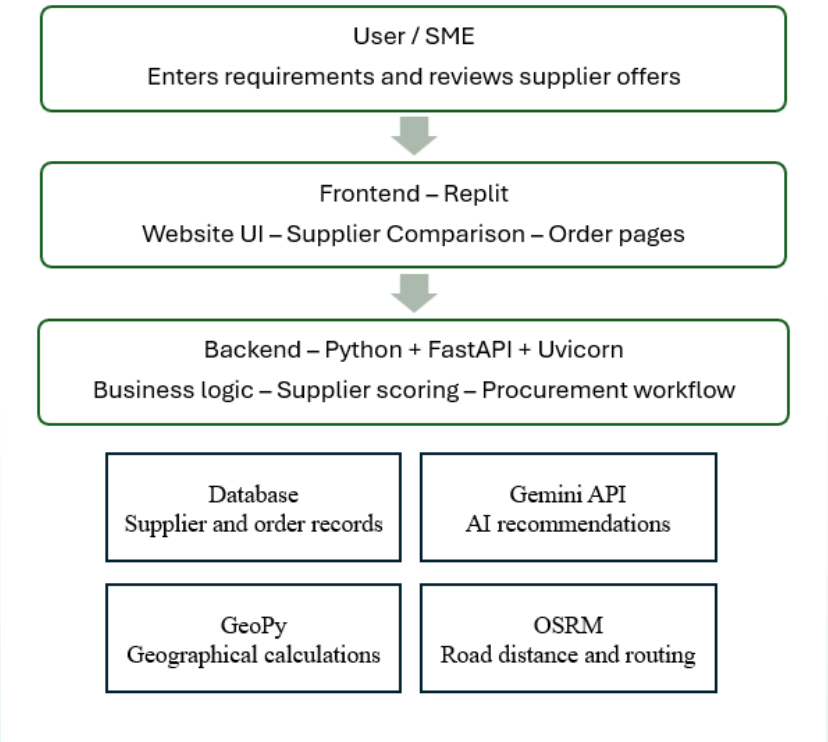
AI Integration: Google Gemini API is intended to generate intelligent supplier recommendations and explain the advantages and trade-offs of different suppliers.

Geographical Services: GeoPy supports geographical calculations, while OSRM can provide road-distance and routing information to help evaluate supplier proximity.

Data Management: A database stores supplier information, product details, purchasing requirements, and order records. The specific database technology will be documented according to the implementation.

System Workflow: The Replit frontend sends procurement requirements to the Python FastAPI backend. The backend retrieves supplier data, calculates comparison scores, and uses geographical information to rank suitable suppliers. Gemini can provide recommendation explanations where integrated. Results are returned to the frontend for users to review and select a supplier. Purchase requests then proceed through human approval, with order outcomes recorded by the backend.

SDC Sponsor-Provided Resources: We will identify any officially provided GenAI API or cloud platform used by the team and describe its role. Any resources not provided by the SDC sponsors will be identified separately.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied a problem-oriented design-thinking approach by discussing common procurement challenges faced by SMEs, particularly the time and effort required to compare supplier prices, shipping fees, minimum order quantities, and delivery times. Although we have not yet conducted direct user interviews, these discussions helped us identify key pain points and define our solution requirements.

We developed a working prototype with a user-friendly interface and an automated supplier comparison workflow. The frontend is built using Replit, while Python and FastAPI power the backend. Internal team testing has been conducted to check the main workflow. Gemini API, OSRM, and GeoPy are intended to support AI-generated recommendations and geographical calculations which are subject to further integration and validation.

In phase 2, we plan to expand the supplier database beyond stationery to cover more product categories and extend geographical coverage beyond Perak. We also aim to introduce a supplier-side interface with purchase orders and order management features, alongside invoices and allow users to select their preferred shipping

companies. Further testing and SME user feedback will guide improvements to usability, recommendation accuracy, and system reliability before wider deployment.

This iterative approach aims to develop a scalable, practical procurement platform that supports SME efficiency and stronger business partnerships.

(200 words)

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.3 – Promote development-oriented policies that support productive activities, entrepreneurship, innovation, and the growth of SMEs SDG 9.4 – Upgrade infrastructure and industries to make them more sustainable through improved resource efficiency and the adoption of clean and environmentally sound technologies SDG 12.2 – Achieve the sustainable management and efficient use of natural resources by encouraging SMEs to make informed purchasing decisions and avoid unnecessary procurement SDG 17 – Encourage and promote effective partnerships between SMEs and suppliers

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	SMEs often rely on WhatsApp, phone calls, and messages to compare wholesale suppliers. Example: for a request of 100 rulers with a MYR4.00 budget delivered to Kuala Lumpur, our prototype returned four ranked supplier offers in under five minutes. However, the manual comparison time has not yet been measured, so further testing is needed to quantify the time saved compared with traditional procurement methods.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% reduction in procurement time reduction, potential cost savings, or supplier comparison efficiency
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve at least a 30% reduction in supplier comparison time and help SMEs identify more cost-effective suppliers through transparent comparisons of prices, shipping fees, minimum order quantities, and delivery times. These targets will be evaluated through prototype testing against manual procurement, while cost savings will be measured using comparable supplier quotations.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type (code/dataset/model)**	**License**
FastAPI	Open-source framework	MIT
Uvicorn	Open-source library	BSD-3-Clause
GeoPy	Open-source library	MIT
OSRM (Open Source Routing Machine)	Open-source routing engine	BSD-2-Clause
Google Gemini	Pre-trained AI model	Google's applicable service terms
Python	Programming language	PSF License
Replit	Development platform	Proprietary service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

[Date] 24th October 2026